

Experiment - 14.

Aim: To design, simplify and implement a simple 4-state traffic light controller using a 2-bit counter (flip-flops) and combinational decoding logic and verify the timing sequence using LEDs on a breadboard.

Apparatus Required

- Breadboard
- IC 7476 / 7474 (Flip-flops)
- IC 7400 / 7408 / 7432 (Logic gates)
- Clock pulse generator / 555 timer
- LEDs (Red, Yellow, Green)
- Resistors (220 Ω)
- Connecting wires
- Power supply (5V)

Theory

A traffic light controller operates in a sequence of states.

Here, a 2-bit counter is used, which gives 4 states (00, 01, 10, 11)

Each state corresponds to a traffic condition:

state (Q1 Q0)	Traffic Light
00	Green ON
01	Yellow ON
10	Red ON
11	Red + Yellow ON (transition)

Flip-flops generate the binary sequence, and combinational logic decodes these outputs to control LEDs.

Truth Tables

Q1	Q0	Green	Yellow	Red
0	0	1	0	0
0	1	0	1	0
1	0	0	0	1
1	1	0	1	1

Logic Equations.

From truth tables:

$$\cdot \text{Green} = Q1' \cdot Q0'$$

$$\cdot \text{Yellow} = Q0$$

$$\cdot \text{Red} = Q1$$

Procedure

1. Place ICs on the breadboard.
2. Connect flip-flops to form a 2-bit counter.
3. Provide a clock signal using a pulse generator or 555 timer.
4. Connect outputs (Q1, Q0) to logic gates for decoding.
5. Connect ~~output~~ LEDs with resistors to outputs:
 - Green $\rightarrow Q1'Q0'$
 - Yellow $\rightarrow Q0$
 - Red $\rightarrow Q1$
6. Power the circuit with 5V supply.
7. Observe LED sequence for each clock pulse.

Observation:

- LEDs glow in a cyclic sequence: Green $\rightarrow$ Yellow $\rightarrow$ Red $\rightarrow$ Red + Yellow $\rightarrow$ Repeat

- Each state changes with one clock pulse.

Result:

~~The traffic~~ The 4-state traffic light controller was successfully designed and implemented using a 2-bit counter and logic gates. The output sequence was verified using LEDs.